

M.J. Biopharm Private Limited			
Plot No. 18, International Biotech Park, Hinjawadi, Phase II, Pune 411057.			
Investigation Report			
Date:	26/03/2026	Deviation No.	DEV/QC/25/010
Investigation tool used: ☐ 6M ☒ 5 Why ☐ Brainstorming			
Other Tools (If any):			
Not Applicable			
Details Investigation:			
Define:			
The following checks shall be considered while writing the “Define” section. 1. Clearly define what happens actually. 2. Explain what is different than expected. 3. Mention the location at where the deviation has occurred. 4. Date/time of deviation occurrence and date/time of detection. 5. Mention the name of personnel who is involved in the deviation. 6. Mention initial scope of deviation (i.e., impacted product/Material/Equipment/System/Batches/etc.) While performing the system suitability test (SST) of TOC Analyzer, Instrument ID. I/QC/018, on 21/03/2026 at 13:00, at Instrument Lab - II (FF-16), analyst (Emp.ID. 615) observed that, obtained system suitability test (SST) results i.e. 115.1 % (Recovery of Benzoquinone) not within the acceptance criteria as per SOP/DP/QC/045. (acceptance criteria: 85 % to 115 %). As per SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L_{CPH} with LabSolutions Software, Section No.: 7.4.3.26, The results of system suitability test must not be less than 85 % & must not more than 115 %).			

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**Investigation Report****Details Investigation:****Measure:**

The following checks shall be considered while writing the “Measure” section.

1. Does the summary provide relevant facts and data/information that was reviewed including: environment, process/product history, personnel info (title and job title), controls/control limits, etc.
2. Is a summary of the analysis of the factors and data provided?
3. Is a conclusion statement of the analysis and review provided?
4. If there were Regulatory Notification, were details provided?
5. Is the report written in a logical flow and easily understood by the reader?

Monthly calibration of TOC Analyzer (Instrument ID: I/QC/018) was due on 21/03/2026, accordingly calibration of the TOC Analyzer, planned on 21/03/2026 and work allotted to the analyst (Emp. ID: 615).

As per SOP ‘Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L CPH with LabSolutions software’ (SOP/DP/QC/045-R02), 2- point calibration & System suitability test needs to be performed during monthly calibration.

Therefore, as per plan, prior to initiation of calibration, analyst (Emp. ID: 615) has assigned A.R. No. NR/25/0438 for the calibration activity in accordance with SOP No. SOP/DP/QC/008 and verified all the pre requisites before initiation of 2-Point calibration and System suitability test as per SOP, ‘Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With LabSolutions Software, SOP No.: SOP/DP/QC/045.

As per SOP, analyst injected the blank water to check the TOC, and obtained result (79.78 ppb) was found well within the defined acceptance criteria. The SOP specifies that the TOC of blank water should be Not More Than 100 ppb. Refer attachment I.

Since, TOC of blank water, found within the limit, therefore analyst performed the 2-point calibration to generate the calibration curve and details of the generated curve as below; Refer Attachment II

Name of Curve: CAL_21032026.2026_03_21_11_54_09.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	5.432	0.1581	2.91 %
Two Point Calibration (500 ppb)	42.58	0.4564	1.07 %

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Further calculated the TOC of blank water which is injected to establish the calibration curve and observed that obtained TOC of blank water is 73 ppb, which was within the acceptance criteria (NMT 100 ppb).

Since, obtained results for two-point calibration found well within the limit as per SOP/DP/QC/045, hence analyst proceeded for 'System Suitability Test' (SST), however after completion of the SST, it was observed that obtained SST results i.e. 115.1 % does not meet the acceptance criteria (Limit: 85 % to 115 %). Refer attachment -III).

As per SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L_{CPH} with LabSolutions Software, Section No.: 7.4.3.26, The results of system suitability test must not be less than 85 % & must not more than 115 %, therefore, initiated the deviation for further investigation.

Based on the occurred discrepancy following immediate action has been taken,

1. The ongoing calibration activity was immediately stopped after observing that the SST result (% Recovery of Benzoquinone – 115.1%) was outside the acceptance criteria (85.0%–115.0%).
2. All glassware which are used for blank and preparation of standard solutions were rechecked and quarantined the analysis tray.
3. The "Out of Calibration" Label was affixed on instrument as per SOP/DP/QA/020.
4. Verified the 2-point calibration result and noted that result is well within acceptance criteria.
5. As the System Suitability Test (SST) results did not meet the acceptance criteria, therefore as an immediate action, it was informed to perform the alternative test i.e. 'Oxidizable Substance' as per purified water standard test procedure i.e. DP/STP/UT/0003 for the scheduled water sample analysis.
6. The observed Non-Conformance immediately informed to Team lead and Head QC.

Based on the reported nonconformance, verified the analyst workbench for standard preparation and noted that analyst prepared the sucrose stock standard solution (50000 ppb) in 100 ml volumetric flask and 1,4-Benzoquinone stock standard solution (50000 ppb) in 100 ml volumetric flask. Analyst used amber colored volumetric flask during preparation of 1,4-Benzoquinone standard.

The prepared standard solutions for 2-point calibration were verified and noted that Sucrose stock standard solution (50,000 ppb) was prepared in a 100 mL volumetric flask correctly as per the procedure defined in SOP/DP/QC/045.

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The calibration data sheet (Format No. SOP/DP/QC/045/F01 and SOP/DP/QC/045/F02) was verified for the recorded preparation details of Sucrose and 1,4-Benzoquinone standard solutions and noted that, analyst prepared the solutions correctly. (Refer attachment -IV & V), the details are as below;

Preparation of Sucrose Standard Solution:

Weighed 11.883 mg of sucrose & transferred in 100 ml volumetric flask and diluted to 100 ml with purified water. (Refer Attachment IV & VI)

Further, taken 1 ml of above stock standard (50000 ppb) by pipetting with Glass pipette in 100 ml volumetric flask and make up the final volume to 100 ml with purified water. (Refer Attachment IV)

Verified the details of sucrose standard which was used for the preparation of 500 ppb solution and noted that analyst used the correct and valid standard, the details are as below;

~~Name of Standard: Sucrose~~

Lot No: 4178807, (In-house No.: PS/009/001)

Validity: 17/02/2027

Preparation of 1, 4-Benzoquinone Standard Solution:

Weighed and dissolved 7.499 mg of 1, 4-Benzoquinone USPRS in 100 ml of purified water. (Refer Attachment V & VI)

Further, diluted 1 ml of 1, 4-Benzoquinone stock standard solution (50000 ppb) to 100 ml with purified water to get 500 ppb 1, 4-Benzoquinone.

Verified the details of 1,4 –Benzoquinone standard which is used for the preparation of 500 ppb solution and noted that analyst had used the correct and valid standard, the details are as below; (Refer Attachment V)

Name of Standard: 1,4 –Benzoquinone USPRS

Lot No: R15690 (In house Vial No.: RS/002/23-002)

Validity: Current lot

After preparation of standards analyst performed the 2-point calibration by preparing the calibration curve 'CAL_21032026' by referring the SOP/DP/QC/045 and generated curve 'CAL_21032026.2026_03_21_11_54_09.cald'. Obtained results of the 2-point calibration found well within the acceptance criteria, the details are as follow, (Refer Attachment - II)

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After generation of the calibration curve, performed the SST by selecting the latest calibration curve 'CAL_21032026.2026_03_21_11_54_09.cald', however after completion of the SST, it was observed that obtained SST results i.e. 115.1 % does not meet the acceptance criteria (Limit: 85 % to 115 %). Refer attachment -III).

SST Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
System Suitability Test'	48.19	1.227	2.55 %
Recovery of 1, 4-Benzoquinone (Limit: Between 85-115 %)	115.1 %		

Verified the vial of 1,4 –Benzoquinone USPRS, Lot No: R15690 (In house Vial No.: RS/002/23-002) to check any abnormality with the remaining quantity of standard and noted that, on document remaining quantity was 41.988 mg, however physically the standard quantity almost consumed as it was lost during the handling and remaining quantity not sufficient for further activity.

Since, assignable cause could not be identified during the preliminary investigation, therefore conducted the analyst interview to find out the root cause and as an outcome of interview, it was suspected that the non-conformance was occurred might be due to dilution error which may have occurred inadvertently while preparing the standard solution, leading to a slightly higher recovery of the 1, 4-Benzoquinone. Refer attachment VII (Analyst interview).

Based on the outcome of preliminary investigation and analyst interview, assignable cause could not be identified, however suspected cause for the occurred non-conformance could be dilution error which may have occurred inadvertently while preparing the standard solution, leading to a slightly higher recovery of the 1, 4-Benzoquinone, therefore to find out the root cause additional investigation followed by hypothesis testing performed as per below recommended actions.

Recommended action I:

Based on the suspected cause, higher side results might be occurred due to improper dilution of the 1,4 – Benzoquinone standard. Hence, to rule out the dilution error it is recommended to perform the re-dilution of 1,4-Benzoquinone stock standard solution (50000 ppb) as below and inject the solution as per below sequence.

Take 1 ml of 1,4-Benzoquinone stock standard solution (50000 ppb) and dilute to 100 ml with purified water.
 (1,4 -Benzoquinone Working Standard (500 ppb);

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Sr. No.	Name of Sample	No. of Sample to be injected
01	Same Vial	01
02	Refilled Vial	01
03	Re-Diluted Vial	01

Outcome of Recommended Action I:

As per recommended action – I, performed the re-dilution of 1,4-Benzoquinone stock standard solution (50000 ppb) as below;

Re-Dilution: Taken 1 ml of 1,4 - Benzoquinone stock standard solution (50000 ppb) and diluted to 100 ml with purified water. (1,4 - Benzoquinone Working Standard (500 ppb).

Further, filled the Re-diluted standard solution and existing standard solution of 1,4 - Benzoquinone standard solutions (500 ppb) in two individual 40 ml cleaned, dried TOC vial and kept vial in TOC autosampler tray

As per the recommended action I, analyst injected 1,4 - Benzoquinone standard solutions (500 ppb) as per below sequence in the TOC analyzer to evaluate the potential dilution error and to ensure the proper functioning of instrument. The analysis was performed following all applicable procedural requirements as per SOP/DP/QC/045 by selecting the latest calibration curve 'CAL_21032026.2026_03_21_11_54_09.cald'

Sr. No.	Name of Sample	No. of Sample to be injected
01	Same Vial	01
02	Refilled Vial	01
03	Re-Diluted Vial	01

After completion of above run, verified the obtained results and noted that SST results obtained from same vial found comparable with the initially obtained results and results obtained from 'Refilled Vial' found slightly lower than the initial results, however results obtained from 'Re-Diluted Vial' found at lower side significantly. Refer Attachment IX, X and XI.

Name of Vials	Hypothesis Testing - System Suitability Test Results			
	Recovery of 1,4-Benzoquinone	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Same Vial	117.0 %	48.91	0.6486	1.33 %
Refilled Vial	104.6 %	44.27	0.9340	2.11 %
Re-Diluted Vial	76.26 %	33.76	0.3251	0.96 %
(SST Limit: Between 85-115 %)				

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Based on the outcome of recommended action – I, it is observed that system suitability results obtained from ‘Same vial’ does not meet the acceptance (Limit: 85 % to 115 %). However, results obtained from ‘Refill vial’ meet the acceptance and there was variability in obtained results as SST results obtained from ‘Re-diluted vial’ found very less as compare with the initial SST results and it might be happened probably due to uneven mixing or improper preparation of the standard solution. Hence to rule out the preparation error, additional investigation followed by hypothesis testing carried out as per below recommended action.

Recommended action II:

Based on the probable cause, it is recommended to prepare the fresh solution of 1,4 – Benzoquinone by taking the new vial of 1,4 –Benzoquinone USPRS, Lot No: R15690 (In house Vial No.: RS/002/23-003) as below under the supervision of QC reviewer.

Preparation of Benzoquinone stock standard solution (50000 ppb):

Weigh 7.5 mg of 1,4 -Benzoquinone & transfer in 100 ml volumetric flask, then add 50 ml of purified water in above 100 ml volumetric flask, sonicate for 30 seconds to dissolve completely and then make up the final volume to 100 ml.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Take 1 ml of above stock standard (50000 ppb) by pipetting with Micropipette/ Glass pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with TOC water.

Inject the freshly prepared 1,4 -Benzoquinone (500 ppb) solution as per below sequence.

Sr. No.	Name of Sample	No. of Sample to be injected
01	1,4 -Benzoquinone	01

Outcome of Recommended Action II:

As per recommended action, hypothesis activity planned on same day i.e. 21/03/2026 and work allotted to analyst Emp. Id. 952 as earlier involved analyst Emp. Id. 615 was left for the day.

As per plan analyst Emp. Id. 952 verified all the pre requisites before initiation of preparation for System suitability test as per SOP, ‘Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With Lab Solution Software, SOP No.: SOP/DP/QC/045.

After ensuring the availability of all the prerequisites, Analyst taken the new vial of 1,4 –Benzoquinone USPRS, Lot No: R15690 (In house Vial No.: RS/002/23-003) and initiated the preparation of 1,4 – Benzoquinone standard, under the supervision of QC reviewer Emp.ID 616 and the details are as below;

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Preparation of Benzoquinone stock standard solution (50000 ppb):

Weighed 7.508 mg of 1,4 -Benzoquinone & transferred in 100 ml volumetric flask, then added 50 ml of purified water in above 100 ml volumetric flask, sonicated for 30 seconds to dissolve completely and then make up the final volume to 100 ml.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Taken 1 ml of above stock standard (50000 ppb) by pipetting with 1 ml glass bulb pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with ultra-pure water.

Further, filled the freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) in 40 ml cleaned and dried TOC vial and kept vial in TOC autosampler tray.

As per the recommended action, injected the freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) as per below sequence in the TOC analyzer to evaluate the potential error with respect to uneven mixing or improper preparation of the standard solution and the activity was performed by following all applicable procedural requirements as per SOP/DP/QC/045 by selecting the latest calibration curve 'CAL_21032026.2026_03_21_11_54_09.cald'

Sr. No.	Name of Sample	No. of Sample to be injected
01	1,4 -Benzoquinone	01

After completion of run, verified obtained results and noted that SST results obtained from freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) i.e. 79.76 % does not meet the acceptance (Limit: 85 % to 115 %), however obtained results are comparable with the SST results i.e. 76.26 % (Mean Area - 33.76, SD 0.3251 and CV 0.96 %) which was obtained from 'Re-Diluted Vial' during the earlier hypothesis testing.

Name of Vials	Hypothesis Testing - System Suitability Test Results			
	Recovery of 1,4-Benzoquinone	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Freshly prepared 1,4 - Benzoquinone	79.76 %	35.06	0.1652	0.47 %

(SST Limit: Between 85-115 %)

Mean area of freshly prepared 1,4 – Benzoquinone (35.06) and mean area obtained from 'Re-Diluted Vial' (33.76) found at lower side as compare with the mean area of 1,4 – Benzoquinone obtained during initial analysis (48.19), hence verified the last calibration report to check the mean area of 1,4 – Benzoquinone and noted that, % recovery and mean area of 1,4 – Benzoquinone obtained during last calibration which was

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performed on 12/03/2026 was 98.44 % and 36.45 respectively and it is comparable with the mean area of freshly prepared 1,4 – Benzoquinone (35.06).

Since, mean area of 1,4 – Benzoquinone obtained during last calibration and current calibration from the freshly prepared 1,4 – Benzoquinone are comparable, however there was huge variation in the % recovery, as the % recovery obtained during last calibration was 98.44 % and in the current calibration it is 79.76 %.

Calculation of the % recovery dependent on the mean area of water (0 ppb) and Sucrose standard (500 ppb) obtained during the 2-Point calibration, hence verified the details of 2-Point calibration from the current calibration and last calibration, and details are as below;

Current Calibration: (Date: 21/03/2026)

Name of Curve: CAL_21032026.2026_03_21_11_54_09.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	5.432	0.1581	2.91 %
Two Point Calibration (500 ppb)	42.58	0.4564	1.07 %

Last Calibration: (Date: 12/03/2026)

Name of Curve: CAL_12032026 01.2026_03_12_15_48_20.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	3.153	0.2727	8.65 %
Two Point Calibration (500 ppb)	36.98	0.2905	0.79 %

Based on the data evaluation of 2-Point calibration from the current calibration and last calibration, it was observed that, during the last calibration, mean area of Sucrose (36.98) (Refer Attachment-XIV) and 1,4 – Benzoquinone (36.45) (Refer Attachment-XV) found comparable with each other, therefore % recovery obtained during last calibration was 98.44 %, however there is variation in the mean area of Sucrose (42.58) and freshly prepared 1,4 – Benzoquinone (35.06) obtained during current calibration, which led to % recovery obtained during current calibration was 79.76 %.

Based on the evaluation of data and obtained results, it is observed that mean area of 1,4 – Benzoquinone (36.45) obtained during last calibration and mean area of freshly prepared 1,4 – Benzoquinone (35.06)

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obtained during latest calibration, mean area obtained from 'Re-Diluted Vial' (33.76) are found comparable with each other, however mean area obtained during initial run of current calibration (48.19), Same vial (48.91), Refilled Vial (44.27) are found varies from the mean area of freshly prepared 1,4 – Benzoquinone and area obtained during latest calibration.

Based on the outcome of above investigation and obtained results, assignable cause for occurred nonconformance could not be identified, however most probable cause for occurrence variation in the mean area of sucrose and 1,4 – Benzoquinone could be uneven mixing or improper preparation of the standard solution, therefore 2-Point calibration to be repeated as the latest generated calibration curve could not be used for the system suitability test and repeat calibration shall be performed as per below recommendation.

Recommended action III:

Based on the most probable cause, it is recommended to repeat the 2-Point calibration and system suitability test by using freshly prepared Sucrose and 1,4 – Benzoquinone standard solution as per below procedure under the supervision of QC reviewer.

Note: Newly generated calibration curve shall be used for the system suitability test.

Sucrose stock solution (50000 ppb):

Weigh accurately 11.88 mg of sucrose and transfer into the 100 mL volumetric flask which containing purified water, shake well and sonicate for 30 seconds to dissolve completely and dilute to 100 mL with purified water.

Sucrose standard solution (500 ppb):

Take 1 ml of above stock standard (50000 ppb) by pipetting with Micropipette / Glass pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with Purified water.

Preparation of Benzoquinone stock standard solution (50000 ppb):

Weigh 7.5 mg of 1,4 -Benzoquinone & transfer in 100 ml volumetric flask, then add 50 ml of purified water in above 100 ml volumetric flask, sonicate for 30 seconds to dissolve completely and then make up the final volume to 100 ml.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Take 1 ml of above stock standard (50000 ppb) by pipetting with Micropipette/ Glass pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with TOC water.

Inject the freshly prepared 1,4 -Benzoquinone (500 ppb) solution as per below sequence.

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Sr. No.	Name of Sample	No. of Sample to be injected
01	Sucrose (0 ppb and 500 ppb)	01
02	1,4 -Benzoquinone	01

Outcome of Recommended Action III:

As per recommended action, repeat calibration activity planned on same day i.e. 21/03/2026 and work allotted to analyst Emp. Id. 952.

As per plan analyst Emp. Id. 952 verified all the pre requisites before initiation of preparation for 2-point calibration and System suitability test as per SOP, 'Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With Lab Solution Software, SOP No.: SOP/DP/QC/045.

After ensuring the availability of all prerequisites, Analyst initiated the preparation of Sucrose and 1,4 – Benzoquinone standard solution, under supervision of QC reviewer Emp.ID 616 and details are as below;

Sucrose stock solution (50000 ppb):

Weighed accurately 11.892 mg of sucrose and transferred into the 100 mL volumetric flask which containing purified water, shake well and sonicated for 30 seconds to dissolve completely and diluted to 100 mL with purified water. Refer Attachment XVII & XVIII.

Sucrose standard solution (500 ppb):

Taken 1 ml of above stock standard (50000 ppb) by pipetting with 1 ml glass bulb pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with Purified water.

After completion of preparation, analyst filled the freshly prepared Sucrose standard solutions (500 ppb) and water in two individual 40 ml cleaned and dried TOC vial and kept vial in TOC autosampler tray.

Further, initiated the run for 2-point calibration to generate the calibration curve and details of the generated curve as below; Refer Attachment XIX.

Name of Curve: CAL_21032026_01.2026_03_21_20_39_03.cald

Calibration Parameter	Calibration Date: 21/03/2026		
	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Two Point Calibration (0 ppb)	4.431	0.08021	1.81 %
Two Point Calibration (500 ppb)	36.82	0.8698	2.36 %

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Further calculated the TOC of blank water which is injected to establish the calibration curve and observed that obtained TOC of blank water is 68 ppb, which was within the acceptance criteria (NMT 100 ppb).

Since, obtained results for two-point calibration found well within the limit as per SOP/DP/QC/045, therefore analyst prepared the 1,4 -Benzoquinone standard solution and proceeded for 'System Suitability Test' (SST), the details of preparation are as below;

Preparation of Benzoquinone stock standard solution (50000 ppb):

Weighed 7.510 mg of 1,4 -Benzoquinone & transferred in 100 ml volumetric flask, then added 50 ml of purified water in above 100 ml volumetric flask, sonicated for 30 seconds to dissolve completely, then make up the final volume to 100 ml. Refer Attachment XVII & XVIII.

Preparation of 1,4 -Benzoquinone Working Standard (500 ppb):

Taken 1 ml of above stock standard (50000 ppb) by pipetting with 1 ml glass bulb pipette in 100 ml in a volumetric flask and make up the final volume to 100 ml with ultra-pure water.

Further, filled the freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) in 40 ml cleaned and dried TOC vial and kept vial in TOC autosampler tray and entire activity was performed by following all applicable procedural requirements as per SOP/DP/QC/045 by selecting the latest calibration curve 'CAL_21032026_01.2026_03_21_20_39_03.cald'.

Sr. No.	Name of Sample	No. of Sample to be injected
01	1,4 -Benzoquinone	01

After completion of above run, verified the obtained results and noted that SST results obtained from freshly prepared 1,4 - Benzoquinone standard solutions (500 ppb) i.e. 98.73 % are found well within the acceptance criteria, the details are as below (Refer Attachment XX).

Name of Vials	Hypothesis Testing - System Suitability Test Results			
	Recovery of 1,4-Benzoquinone	Mean Area	SD (NMT 0.5)	CV (NMT 5.0 %)
Freshly prepared 1,4 - Benzoquinone	98.73 %	36.41	0.5314	1.46 %
(SST Limit: Between 85-115 %)				

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Based on the obtained results, it is observed that mean area sucrose (36.82) obtained from 2-point calibration and mean area of 1,4 – Benzoquinone (36.41) obtained from system suitability run found comparable with each other which led to % recovery found well within the acceptance criteria.

Based on the outcome of investigation and obtained results, it was confirmed that, during the initial calibration, sucrose and 1,4 – Benzoquinone standard solutions were prepared improperly or solutions were not mixed uniformly which led to variation in the area of sucrose and 1,4 – Benzoquinone consequently % recovery of the 1,4 – Benzoquinone found at higher side, however during the repeat calibration analyst prepared the standard solution properly by sonication (30 Seconds) and as an outcome all the obtained results found well within the acceptance criteria.

As a part of investigation, operation procedure 'Title: Operation, Calibration and Maintenance of Shimadzu TOC Analyzer Model No.: TOC-L CPH With Lab Solution Software, SOP No.: SOP/DP/QC/045 of TOC analyzer was verified to check the instructions related to sonicate the standard solution to dissolve the sucrose and 1,4 – Benzoquinone standard completely to get the uniformly mixed solution, however upon verification it was noted that SOP doesn't have clarity to sonicate the standard solution which could be the contributing factor for the reported nonconformance.

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Analyze: Various investigation tools used to analyze the root cause like 6M, 5 Why, Personnel interview, Brainstorming etc.

6 M Method (If Applicable):

1. **Man:**
Not Applicable.
2. **Machine:**
Not Applicable.
3. **Measurement:**
Not Applicable.
4. **Material:**
Not Applicable.
5. **Method:**
Not Applicable.
6. **Milieu (Environment):**
Not Applicable.

Conclusion:

Not Applicable.

Note: No any 6 M question shall be deleted in the investigation, if any question is not applicable, then mention in ans. as "Not Applicable".

5 Why Approach (If Applicable):

1. WHY: Why did the System suitability test fail?

Ans. The system suitability failure occurred because the % recovery of 1,4-Benzoquinone exceeded the acceptance criteria.

2. WHY: Why did the % recovery exceed the acceptance limit?

Ans. The % recovery exceeded the acceptance limit because the system response was elevated, leading to higher blank and recovery readings.

3. WHY: Why was the system response elevated?

Ans. The system response was elevated because of improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution.

4. WHY: Why was the preparation of 1,4-Benzoquinone standard solution was improper/ unevenly mixed?

Ans. Preparation of 1,4-Benzoquinone standard solution was improper/ unevenly mixed because it was not sonicated.

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**Investigation Report****5. WHY: Why was the 1,4-Benzoquinone standard solution not sonicated?**

Ans. 1,4-Benzoquinone standard solution not sonicated because guidance procedure doesn't have clarity to sonicate the standard solution and due to Lack of clarity in the SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L CPH with LabSolutions Software, 1,4-Benzoquinone and sucrose standard solution were not sonicated.

***6. WHY: Not Applicable**

Ans. Not Applicable

Conclusion: Based on the 5 Why Approach outcome, it is concluded that nonconformance was occurred due to improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure, as the SOP doesn't have clarity to sonicate the standard solution and due to Lack of clarity in the SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L CPH with LabSolutions Software, 1,4-Benzoquinone and sucrose standard solution were not sonicated.

*Note: Less or more than 5 questions may be asked based on the outcome of earlier questions.

Brainstorming:

Not Applicable

Other Tool if Any:

Not Applicable

Other Tool if Any:

Not Applicable

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Investigation Report

Investigation Outcome:

The investigation has been carried out by using the '5 Why' investigation tool. The investigation was driven through the DMAIC methodology.

Based on the initial risk assessment, the deviation has been categorized the deviation as 'Major'.

The nonconformance occurred during the calibration of TOC Analyzer (Instrument ID: I/QC/018) for the system suitability test parameter at Instrument lab-II (FF-16), as the analyst (Emp. ID: 615) on 21/03/2026 observed that the system suitability test (SST) of TOC Analyzer (Recovery of Benzoquinone) not within the acceptance criteria i.e. 115.1 % as per SOP/DP/QC/045. (Limit: 85 % to 115 %) therefore performed the detailed investigation for SST failure followed by multiple hypothesis testing and as an outcome it was revealed that nonconformance was occurred due to improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure, as the SOP doesn't have clarity to sonicate the standard solution and due to Lack of clarity in the SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L_{CPH} with LabSolutions Software, 1,4-Benzoquinone and sucrose standard solution were not sonicated. Based on the investigation outcome root cause is identified as 'Procedure Error i.e. Lack of clarity in the guidance procedure SOP No.: SOP/DP/QC/045, as the SOP doesn't have clarity to sonicate the standard solution, therefore as a corrective action repeated the calibration by sonicating the 1,4-Benzoquinone and sucrose standard solution and as an outcome calibration results were found stable and within acceptance criteria. Based on the identified root cause, performed the impact assessment found that there is no impact on the previous analysis. The observed SST failure prevented the TOC analysis of the scheduled water sample until the cause was investigated and resolved. As an alternative measure to ensure timely monitoring of water quality, the test for Oxidizable Substances performed and result were well within the acceptance criteria. No product quality impact was anticipated, as the issue not related to sample testing. Also, there is no impact on the performance of the instrument. However, operation procedure of the TOC Analyzer found impacted as the SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L_{CPH} with LabSolutions Software doesn't have clarity to sonicate the standard solution. Based on the nature of nonconformance, chronology of events, and details investigation, further root cause categorization has been done as per format for "Root Cause Categorization" SOP/DP/QA/008-F03. (Refer Attachment No. XXII).

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Identified Root Cause/ Probable Cause:

Based on the detailed investigation, below mention probable root cause were identified

Primary Root Cause Level 1: Procedure.

Secondary Root Cause Level 2: Document unclear/Lacking details.

Third Root Cause Level 3: NA.

Impact Assessment (System/ Document/ Product/ Equipment/Patient safety/Past batches):

System: The non-conformance related to failure of system suitability test (SST) of TOC Analyzer is attributed to the Procedure error as the obtained SST results does not meet the predefined acceptance criteria, however after evaluation, it has been determined that this non-conformance does not have any anticipated impact on the overall system.

Document: Performed the detail impact assessment for the reported nonconformance and as an outcome the root cause identified as 'Procedure error' as elevated SST results were occurred due to improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure, as the SOP doesn't have clarity to sonicate the standard solution and due to Lack of clarity in the SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L_{CPH} with LabSolutions Software, 1,4-Benzoquinone and sucrose standard solution were not sonicated, therefore SOP No.: SOP/DP/QC/045 found impacted and needs to be revised to incorporate the clarity to sonicate the 1,4-Benzoquinone and sucrose standard solution.

Product: The non-conformance related to failure of system suitability test (SST) of TOC Analyzer is attributed to the Procedure error as the obtained SST results does not meet the predefined acceptance criteria, and it was occurred during the scheduled monthly calibration, therefore no any samples were analyzed from the date of occurrence of nonconformance. As an alternative measure to ensure timely monitoring of water quality, the test for Oxidizable Substances performed and result were well within the acceptance criteria. Therefore, impact on product quality not anticipated due to reported nonconformance.

Instrument: Performed the detail impact assessment for the reported nonconformance and as an outcome the root cause identified as 'Procedure error' as elevated SST results were occurred due to improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure. As a corrective action repeated the calibration by sonicating the sucrose and 1,4-Benzoquinone standard solution for 30 seconds, as an outcome the results of 2-point calibration, system suitability was found well within limits, hence there is no impact on the performance of the instrument.

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Investigation Report

Patient safety/Past Batches:

Performed the detail impact assessment for the reported nonconformance and as an outcome the root cause identified as 'Procedure error' as elevated SST results were occurred due to improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure and it did not impact on the patient safety, as the affected calibration curve (CAL_21032026.2026_03_21_11_54_09.cald) was not used for any routine or batch testing. All previously released water samples were tested using valid calibration curve. There were no any out-of-specification results or product quality deviations were reported for previously analyzed samples. Hence there is no impact on the previously performed analysis and quality of 'Product', Patient safety/Past Batches.

Improve: Improve section covers the corrective actions

Following checkpoint shall be considered as guidance only while finalizing the corrective actions,

1. Is a specific corrective Actions identified (including applicable Immediate Actions) to remediate the current issue or situation such that the associated system was returned to a state of control/compliance and also it will avoid the recurrence of deviation?
2. Is specific Corrective Actions identified for each root cause /substantiated probable root cause, as applicable?
3. Is the Corrective action assigned a unique number, responsible person and due date so it can be tracked? Does the action describe what will be the expected outcome they can be verified?
4. Are the identified corrective actions achievable based on the information provided?

Corrective Action:

The investigation has been driven through the DMAIC methodology. During the investigation '5 Why' tool used to find out the root cause and based on the detailed investigation, it has been concluded that the nonconformance was triggered due to 'Procedure error' as elevated SST results were occurred due to improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure, therefore, as a corrective action, repeated the calibration by sonicating the sucrose and 1,4-Benzoquinone standard solution for 30 seconds, as an outcome the results of 2-point calibration and system suitability were found well within acceptance criteria. The previously generated impacted calibration curve (CAL_21032026.2026_03_21_11_54_09.cald) was considered as invalidated, and newly generated calibration curve (CAL_21032026_01.2026_03_21_20_39_03.cald) considered valid. Analyst re-trained and instructed them to prepare the sucrose and 1,4-Benzoquinone by sonication for 30 seconds and to select the of valid calibration curves for routine analysis.

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**Investigation Report****Control: Control section covers the preventive actions****Following checkpoint shall be considered as guidance only,**

1. Is specific Preventive Actions identified for each root cause /substantiated probable cause as applicable?
2. Is the Preventive Action linked the classification of the root cause and explanation given for how it will prevent occurrence?
3. Is the Preventive action assigned a unique number, responsible person and due date so it can be tracked?
Does the action describe and expected outcome can be verified?
4. Is an Interim Plan needed to ensure a state the control while the Preventive Actions were implemented?
5. Is rationale provided when no Preventive Action were identified?
6. Does the Final Comments section include rotational to support the conclusion of the investigation and CAPA
7. Is each of the impact assessment fields completed correctly — Regulatory Impact, Regulatory notification Product Quality, Validation, Stability, Market/Clinical?
8. Does the Recommended Lot disposition match the conclusions of the investigation and Impact assessment?
9. Does the Conclusion include final decision and rationale other regulatory notification required?
10. CAPA required to release material or batches have been verified to be complete and closed prior to material or batch disposition. Any related CAPA that remain open, but have no impact on the material or batch release and a documented explanation is provided.
11. Does the Conclusion include a brief summary of root cause, final scope/impact including rationale for product/material disposition, impact assessment and relevant lot/material/equipment details?
12. Are the identified preventive actions achievable based on the information provided?

Preventive Action:

As an outcome of investigation, it is identified that the nonconformance occurred due to 'Procedure Error i.e. improper preparation/uneven mixing of the 1,4-Benzoquinone standard solution and it was contributed by the lack of clarity in the guidance procedure, as the SOP doesn't have clarity to sonicate the standard solution and due to Lack of clarity in the SOP No.: SOP/DP/QC/045, Revision No: R02, Title: Operation, Calibration, Maintenance of Shimadzu TOC Analyzer Model No. TOC-L CPH with LabSolutions Software, 1,4-Benzoquinone and sucrose standard solution were not sonicated, therefore SOP No.: SOP/DP/QC/045 found impacted and needs to be revised to incorporate the clarity to sonicate the 1,4-Benzoquinone and sucrose standard solution.

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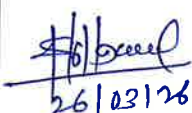
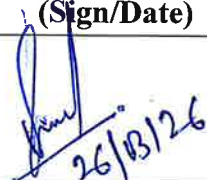
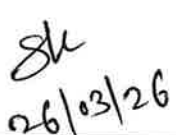
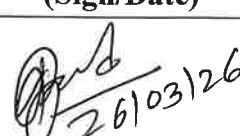
Investigation Report

Document Reviewed:

1. SOP No.: SOP/DP/QC/045

List of attachment (If applicable):

Attachment – I: Initial Blank Water TOC Result
 Attachment – II: 2-Point Calibration Result (Initial)
 Attachment – III: System Suitability Test Result (Initial)
 Attachment – IV: Preparation Record of Sucrose for 2-Point Calibration.
 Attachment – V: Preparation Record 1,4-Benzoquinone
 Attachment – VI: Weight Print Evidence
 Attachment – VII: Analyst Interview
 Attachment – VIII: Preliminary laboratory investigation & Recommended Action (I)
 Attachment – IX: Hypothesis Result (Same Vial)
 Attachment – X: Hypothesis Result (Refilled Vial)
~~Attachment – XI: Hypothesis Result (Re-Diluted Vial)~~
 Attachment – XII: Outcome Recommendation Action (I) & Recommended action II
 Attachment – XIII: Hypothesis Result (Fresh Preparation of 1,4-Benzoquinone)
 Attachment – XIV: 2-Point Calibration Result (Previous Calibration)
 Attachment – XV: System Suitability Test Result (Previous Calibration)
 Attachment – XVI: Outcome Recommendation Action (II) & Recommended action III
 Attachment – XVII: Weight Print Evidence (Repeat Calibration)
 Attachment – XVIII: Preparation Record of Sucrose & 1,4-Benzoquinone for 2-Point Calibration and SST.
 Attachment – XIX: 2-Point Calibration Result (Repeat Calibration)
 Attachment – XX: System Suitability Test Result (Repeat Calibration)
 Attachment – XXI: Outcome Recommendation Action (III)
 Attachment – XXII: Root Cause Categorization Form
 Attachment – XXIII: Training Documentation Form

Prepared By QC (Sign/Date)	Reviewed By QC (Sign/Date)	Reviewed By QC (Sign/Date)	Reviewed By QA (Sign/Date)	Approved By QA (Sign/Date)
 26/03/26	 26/03/26	 26/03/26	 26/03/26	 26/03/26

21-03-2026 13:00:40(+05:30) : Confirm[Akash Ghadage]Other CommentData submitted for review
21-03-2026 14:46:09(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed

Date of creation 2026/03/21 11:10:53
Sample Run TOC_I-QC-018_2026 - 5-558-13 - 210326.tdx
User Akash Ghadage
Object ID 0L-H54436004105-13525E119A4D-0000
System TOC-L + ASI-L
Origin TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald

Sample

Sample Name: Blank
Sample ID: Blank
Object ID 0L-H54436004105-13525E119A4D-0000
Origin: TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald
Status Completed
Chk. Result USP/EP TOC test: Pass
Comment

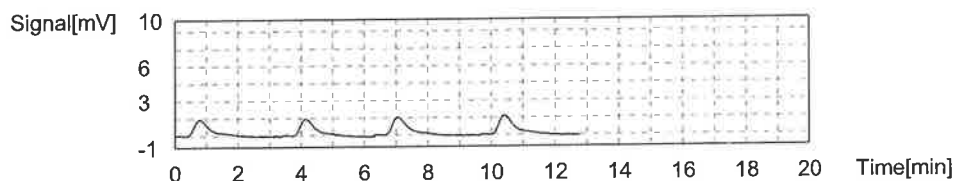
Type	Anal.	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC:79.78ppb

1. Det

Anal.: NPOC

No.	Area	Conc.	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Cal. Curve	Date / Time
1	5.123	75.72ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/21 11:00:15
2	4.610	68.14ppb	816uL	1.000	*****	E	TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/21 11:03:35
3	5.619	83.06ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/21 11:07:21
4	5.450	80.56ppb	816uL	1.000	*****		TOC_I-QC-018_2026 - CAL_12032026_01.2026_03_12_15_48_20.cald	2026/03/21 11:10:52

Acid Add. 0.000%
Spurge Gas Flow 80mL/min
Sp. Time 60.00sec
Mean Area 5.397
Mean Conc. 79.78ppb
SD Area 0.2522
CV Area 4.67%
SD Conc 3.727
CV Conc 4.67%
Vial 4
No Washes 2



History

21-03-2026 13:00:39(+05:30) : Confirm[Akash Ghadage]Other CommentData submitted for review
 21-03-2026 14:46:08(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed

Date of creation 2026/03/21 12:28:11
 Sample Run TOC_I-QC-018_2026 - 5-563-3 - CAL_21032026.tlx
 User Akash Ghadage
 Object ID 0L-H54436004105-13525E11C210-0000
 System TOC-L + ASI-L
 Origin TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald

Cal. Curve

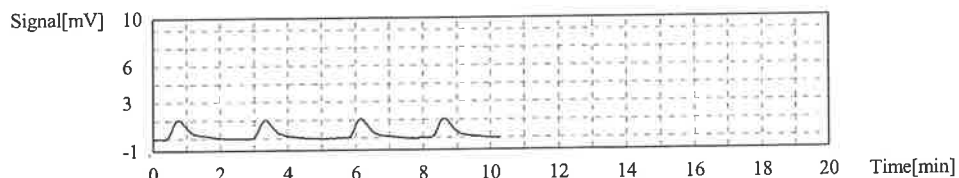
Sample Name: Sucrose
 Sample ID: Standard
 Object ID 0L-H54436004105-13525E11C210-0000
 Cal. Curve: TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald
 Status Completed
 Comment:

Type	Anal.
Standard	NPOC

Conc: 0.000ppb

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	5.603	816uL	1.000	*****		2026/03/21 12:01:41
2	4.460	816uL	1.000	*****	E	2026/03/21 12:04:53
3	5.291	816uL	1.000	*****		2026/03/21 12:07:45
4	5.402	816uL	1.000	*****		2026/03/21 12:10:33

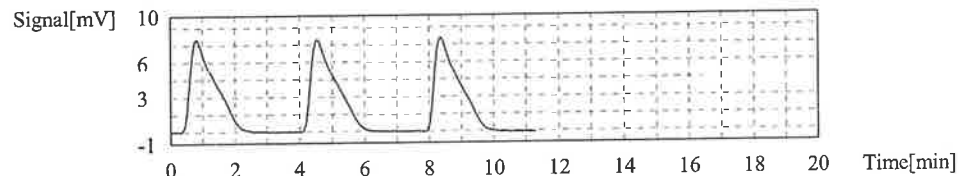
Acid Add. 0.000%
 Spurge Gas Flow 80mL/min
 Sp. Time 60.00sec
 Mean Area 5.432
 SD Area 0.1581
 CV Area 2.91%
 Vial 1
 No Washes 2



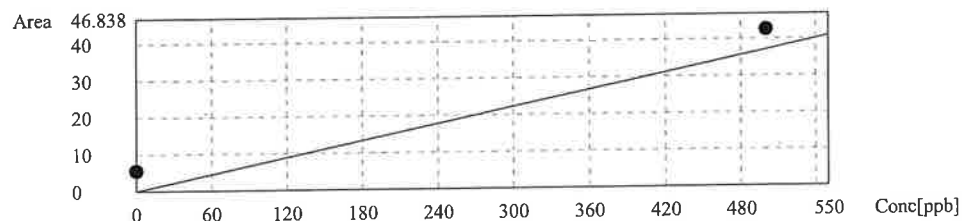
Conc: 500.0ppb

No.	Area	Inj. Vol.	Aut. Dil.	Rem.	Ex.	Date / Time
1	42.07	816uL	1.000	*****		2026/03/21 12:19:50
2	42.95	816uL	1.000	*****		2026/03/21 12:24:02
3	42.72	816uL	1.000	*****		2026/03/21 12:28:09

Acid Add. 0.000%
 Spurge Gas Flow 80mL/min
 Sp. Time 60.00sec
 Mean Area 42.58
 SD Area 0.4564
 CV Area 1.07%
 Vial 2
 No Washes 2



Slope: N/A
 Intercept N/A
 r^2 N/A
 r N/A
 RSE(%)
 Zero Shift Yes



History

No. 1
 Date & Time 2026/03/21 11:50:07(+0530)
 User Name Akash Ghadage
 Changes New
 Contents
 Reason for change

Attachment - II.

21-03-2026 13:00:39(+05:30) : Confirm[Akash Ghadage]Other CommentData submitted for review
21-03-2026 14:46:08(+05:30) : Review[Nilesh Dushing]Other CommentData Reviewed

Date of creation 2026/03/21 12:50:52
Sample Run TOC_I-QC-018_2026 - 5-563-6 - CAL_21032026.tlx
User Akash Ghadage
Object ID 0L-H54436004105-13525E11E152-0000
System TOC-L + ASI-L
Origin TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald

Sample

Sample Name: 1-4 Benzoquinone
Sample ID: std SST
Object ID 0L-H54436004105-13525E11E152-0000
Origin: TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald
Status Completed
Chk. Result 115.1% : USP/EP system suitability test: Fail
Comment

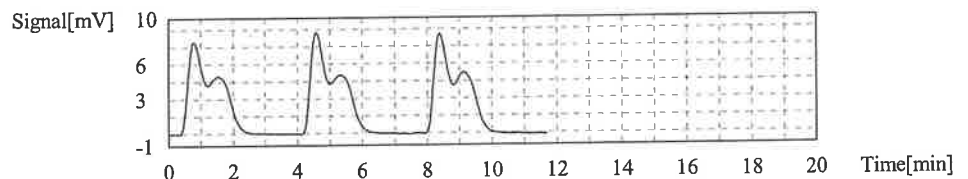
Type	Anal.	Manual Dilution	Result
Unknown	NPOC	1.000	NPOC:648.6ppb

1. Det

Anal.: NPOC

No.	Area	Conc.	Inj. Vol	Aut. Di.	Rem.	Ex.	Cal. Curve	Date / Time
1	46.78	629.6ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 12:42:11
2	49.04	660.1ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 12:46:23
3	48.74	656.0ppb	816uL	1.000	**h****		TOC_I-QC-018_2026 - CAL_21032026.2026_03_21_11_54_09.cald	2026/03/21 12:50:52

Acid Add. 0.000%
Spurge Gas Flow 80mL/min
Sp. Time 60.00sec
Mean Area 48.19
Mean Conc. 648.6ppb
SD Area 1.227
CV Area 2.55%
SD Conc 16.52
CV Conc 2.55%
Vial 3
No Washes 2



History